

**CHARACTERIZING WETLAND CHANGE USING MULTI-DECADAL
LANDSAT IMAGERY: A COMPARISON OF LINEAR TEMPORAL
MIXTURE ANALYSIS, NONLINEAR MANIFOLD MACHINE
LEARNING, AND JOINT CHARACTERIZATION**

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by
Margot M. Mattson
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The Undersigned Faculty Committee Approves the
Thesis of Margot M. Mattson:

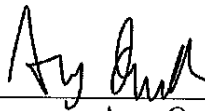
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Trent Biggs (May 8, 2023 14:21 PDT)

Trent Biggs, Co-Chair
Department of Geography


Daniel J. Sousa (May 8, 2023 14:25 PDT)

Daniel Sousa, Co-Chair
Department of Geography



Amy Quandt
Department of Geography



Paul Ganster
Institute for Regional Studies of the Californias

May 8, 2023

Approval Date

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by

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ABSTRACT OF THE THESIS

CHARACTERIZATION OF WETLAND CHANGE USING MULTI-DECADAL LANDSAT IMAGERY: A COMPARISON OF LINEAR TEMPORAL MIXTURE ANALYSIS, NONLINEAR MANIFOLD MACHINE LEARNING, AND JOINT CHARACTERIZATION

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Margot M. Mattson

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Wetlands provide critical habitat for birds and endangered species but have been drastically reduced by changes in water availability and land use. Time series of remote sensing imagery can help monitor wetland and vegetation dynamics, but heterogeneity in trajectories can complicate mapping and visualization of response units that are ecologically meaningful. We used spectral mixture analysis (SMA) on a time series of Landsat imagery (1995-2021) to map changes in surface water and vegetation in the rapidly changing Andrade Mesa wetlands along the US-Mexico border. Three techniques were used to characterize the temporal patterns in the SMA fractions: linear principal component transformation, the nonlinear manifold machine learning algorithm Uniform Manifold Approximation and Projection (UMAP), and Joint Characterization (JC) incorporating both linear and nonlinear techniques. The changes in water and vegetation cover were modeled using temporal mixture analysis and the total loss of surface water and vegetation decline quantified. Linear transformation allowed for the identification of the dominant temporal trends, while UMAP allowed for the separation of distinct trends indistinguishable using PCA alone, but the relationship between the clusters and the represented physical processes was less interpretable. Joint Characterization allowed for improved temporal endmember selection in both the dark and vegetation fraction multi-temporal images compared to PCA and UMAP alone. The methods allowed for the identification of unique trajectories of change associated with different mechanisms and outcomes, including drying of water bodies due to lining of the All-American Canal, mechanical clearing of vegetation, and greening of trees. Surface water and vegetation declined with timing that corresponds with the lining of the AAC, suggesting that this wetland in Mexico is reverting to desert in response to water efficiency measures implemented in the United States. The cross-border environmental and social impacts should be assessed for infrastructure developments in transboundary aquifers to inform mitigation efforts. Future research should explore other potential drivers of wetland decline, such as variation in climate and precipitation, increased groundwater pumping for agriculture in Mexico, and the impact of the regional earthquake in 2010 that disrupted water deliveries through surface canals.

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CHAPTER 1

INTRODUCTION

Wetlands in semi-arid regions are highly important ecosystems that provide critical services such as habitat for migratory birds and endangered species (Minckley et al., 2013), along with provision and cultural services for surrounding communities (Glenn et al., 1999). However, semi-arid wetlands are sensitive to changes in land use, climate variability, and shifts in water allocations due to increased demands for agriculture, industry, and domestic use due to low precipitation rates. The accurate mapping of wetland extent and the characterization of seasonal dynamics is necessary to establish an understanding of wetland hydrology and vegetation coverage that can be used to monitor decline and inform conservation efforts to mitigate lost services caused by habitat degradation.

Multi-spectral repeat satellite imagery has been widely used to monitor changes in semi-arid wetlands (Gxokwe et al., 2020; Ozesmi & Bauer, 2002). However, the use of decameter resolution imagery can prove challenging when documenting changes in seasonal flooding and small waterbodies occurring at subpixel levels. Spectral mixture analysis (SMA) provides a straight-forward method with minimal required pre-processing to accurately estimate the surface water area for small and subpixel waterbodies (Shanmugam et al., 2006), and can be applied to image time series to characterize the hydrological processes of individual wetlands and identify the impacts of climate variability and abnormal changes such as long term decline and changes in water availability with consequences for ecosystem function (Halabisky et al., 2016; Brice et al., 2022; Kissel et al., 2020).

However, while highly useful to analyze changes in hydrographs, fraction water coverage applications in wetlands often rely on user-defined wetland boundaries to conduct the analysis (Halabisky et al., 2016). In complex wetland systems with high spatiotemporal variability, or regions that have already undergone significant habitat degradation, manual delineation of ponds or wetlands is not always practical.

Alternatively, the unique spatiotemporal patterns that represent biophysical processes, such as shifts in surface water and vegetation, can be characterized and modeled at the landscape scale with minimum prior assumptions from the analyst (Small, 2012). Characterization and modeling of high-dimensional remote sensing datasets, such as multi-decadal time series, can be achieved through the use principal component transformations and temporal mixture models (Small, 2012; Sousa & Davis, 2020; Sousa & Small, 2019). Additionally, recent work has shown the ability to characterize long term wetland change trajectories using principal component transformations of multi-temporal spectral indices and machine learning classifiers to create categorical maps of change in different land cover groups from Landsat and MODIS imagery (Kovacs et al., 2022).

More recently, non-linear techniques have been implemented to reduce the dimensionality of datasets and allow for the identification of statistically significant clusters that provide information about functional groups in the landscape (Sousa & Small, 2022a; Bueso et al., 2020; Li et al., 2022). Used together in Joint Characterization (Sousa & Small, 2021), both linear and non-linear approaches can provide complimentary information to characterize and model physical processes represented in the dataset (Sousa & Small, 2022a; Sousa & Small, 2022b; Sousa & Small, 2022c).

In this study, we use a time series of surface water and vegetation fraction estimates generated by SMA to demonstrate the spatiotemporal variability in rapidly changing wetland habitat located in a transnational watershed along the US-Mexico. We explore three questions:

- a) How well do time series of spatially average vegetation and water fractions characterize processes of wetland degradation?
- b) What are the dominant spatiotemporal patterns captured by linear dimensionality reduction (PCA) and non-linear dimensionality reduction (Uniform Manifold Approximation and Projection; UMAP)? To what extent do these patterns complement each other, and in what way can this complementarity be captured by Joint Characterization?
- c) What is the total area of habitat transitions and the implications of wetland habitat loss in a semi-arid, transboundary watershed?

CHAPTER 2

BACKGROUND

2.1 Study Area

The Andrade Mesa wetlands (Fig. 1) are a hydrologically dynamic, ecologically significant Sonoran Desert landscape. Formed on aeolian and alluvial deposits at the base of the Andrade Mesa plateau, the region has a midlatitude desert climate, receiving only 83 mm of rainfall annually from 1969-1999 (CONAGUA, 2020). Temperature ranges from 0-50°C with a median temperature of 22.3°C. The study area is located approximately 30 km east of Mexicali, Baja California, Mexico and approximately 5 km south of the US-Mexico international border.

The soils and sediment which host the wetlands are characterized by high transmissivity and storage potential. The wetlands can be considered a surface expression of the Mexicali Valley aquifer, which in turn is fed by subsurface flow from the Colorado River and the associated irrigation canals. Aquifer recharge has been greatly diminished in recent years due to reduced flows in the Colorado River, extensive groundwater pumping for municipal and agricultural use, and high rates of potential evapotranspiration (approximately 2300 mm per year) (CONAGUA, 2020).